Statistical-mechanical characterization of memory and quench response in state-separated LLM-agent networks

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Abstract

We treat a network of state-separated, locally informed large-language-model (LLM) agent instances as a finite interacting stochastic system. Each persistent identity owns a categorical belief, a committed action, bounded private memory, a local field, an inbox and outbox, and a typed local action interface; a random-sequential scheduler offers update opportunities but does not choose the resulting state. We distinguish the complete augmented simulator state, its observable belief–action projection, and a rolling macroscopic representation. This separation exposes how omitted private memory can act as a hidden historical coordinate. We report collective order, uncertainty, dependence, an effective symmetric-layer energy, fluctuations, and coarse-grained path-reversal divergence. We first correct a derived-analysis error in an earlier Qwen quench experiment without changing any trajectory. We then conduct a prospectively specified two-family experiment with 48 trajectories, 34,560 local decisions, and six independent graph/environment clusters per model. A field reversal and restoration is paired with nominal evolution, genuine persistent memory, and a format-matched scrambled-history control. The independent-family quench effect is 42.263 (95% interval 24.316 to 59.303) within-model distance units. Genuine memory changes bias-adjusted reversal divergence relative to Markovized and scrambled-history controls by 0.05438 (0.03459 to 0.07548) and 0.04845 (0.02190 to 0.07526) nats per attempted update. The fixed early-to-late restoration contrast is 52.541 (35.406 to 68.673) distance units. All four effects meet their prespecified criteria, although the memory contrasts are larger and more uniform in Granite than in Qwen. An out-of-sample kinetic surrogate identifies which response components admit a simple local closure and which remain model- and history-dependent. These finite-size results support a reduced statistical-mechanical description without interpreting decoding temperature as physical temperature, reference energy as literal energy, or projected path divergence as exact thermodynamic entropy production.

Keywords: statistical mechanics of agents; language-model agents; quench dynamics; entropy rate; multi-information; temporal asymmetry.

1 Introduction

Large-language models are increasingly studied not only as isolated prompt–response systems but as components of interacting agent populations [1, 2]. Once agents exchange messages, retain histories, and act through a network, an output from one local decision can become part of another agent's next input. This feedback permits amplification, cascades, coordination, fragmentation, lock-in, delayed response, and incomplete recovery even when every component uses fixed model weights. The capability of one model instance therefore does not determine the trajectory of the group, because repeated circulation can create modes of organization that no single generation exhibits. The scientific question becomes what dynamical organization emerges when many stochastic language-model instances interact over time, a question that matters for the analysis, design, reliability, and control of multi-agent AI systems.

At the component level, an LLM agent places a generative model inside a recurrent loop of observation, decision, action, and environmental feedback. Its future inputs can contain its own earlier outputs, received messages, and changes caused by earlier actions. Population trajectories can consequently depend on message content, network position, update order, retained context, and decoding randomness. Conventional benchmarks can characterize useful local capabilities, but they do not reveal whether network feedback produces stable organization, amplifies a perturbation, stores hidden history, or returns the population to its earlier regime. Those are properties of the interacting process rather than of any response considered alone.

Recent agent architectures establish both the feasibility of such processes and the source of their complexity. Role-playing agents cooperate through natural-language exchange [3], while generative agents combine observation, planning, reflection, and persistent natural-language memory in a simulated social environment [4]. These systems show that retrieved history and communication can materially condition later actions. Yet architecture and application studies are commonly evaluated by task completion or behavioral plausibility. Those objectives do not by themselves specify a controlled stochastic process, distinguish private state from observed state, or identify which population coordinates retain dynamical information.

Computational social simulation and population experiments make the same measurement problem visible from another direction. Conditioned language models reproduce portions of survey-response distributions [5], and simulated samples replicate some established behavioral experiments while exhibiting model-specific distortions [6]; the broader opportunities and validity limits of LLM-empowered agent-based simulation are reviewed in [7]. Networked agents have generated consensus and fragmentation in opinion experiments [8], shared naming conventions and collective bias [9], and model-dependent cooperation and coordination in repeated games [10]. Together, these studies establish that interaction can produce reproducible group-level outcomes, while cautioning against treating semantic plausibility as a dynamical explanation or an LLM population as a faithful proxy for human behavior.

The interaction graph makes collective organization irreducibly networked. Who can influence whom determines the paths along which information and perturbations propagate, while topology, local versus collective information, and community structure can alter the resulting dynamics. Statistical mechanics and network science provide tools for connecting such interaction structure to collective behavior [11, 12]; studies of cooperative processes on networks also emphasize that topology and finite size can qualitatively affect observed regimes [13]. A population mean suppresses this organization: identical mean order can coexist with different fluctuations, graph-distance correlations, community patterns, and routes of influence. Graph distance supplies a natural coordinate for asking whether local perturbations remain localized or become collectively organized. Network-aware observables are therefore needed even when the scientific target is a population-level response.

More broadly, statistical mechanics supplies a disciplined micro-to-macro measurement framework. Agent-based models ask how local rules and heterogeneous interactions generate system-level behavior [14], and statistical physics has developed quantitative accounts of opinion, cultural, language, and consensus dynamics [15, 16]. Canonical stochastic models show how local updates generate collective relaxation [17], how a potential can constrain stationary interaction dynamics [18], and how nonreciprocity can support different collective structures [19]. Local language-conditioned transitions are much more complicated than spin updates, but population order, dependence, correlation, response, and distributions can still be measured. For a finite population observed over finite windows, fluctuations and variation across independent realizations carry information that a mean trajectory discards. This use of statistical mechanics does not require literal particles, a thermodynamic limit, or a phase transition.

Controlled perturbation is important because nominal or steady observations underdetermine dynamics. Similar baseline averages could describe a robustly organized system, a fragile one, a system that responds strongly and recovers, or one with delayed or incomplete recovery. Reversing an external field and then restoring it probes finite-horizon sensitivity, response, relaxation, and recovery directly. Comparing that trajectory with a matched nominal arm also separates intervention-associated change from ordinary evolution over the same observation window. A quench is used here as a dynamical intervention, not as evidence that the network is an equilibrium material.

Persistent history creates a distinct coarse-graining problem. The complete agent process may include private textual state that is absent from a measured belief–action projection, so the observed population trajectory need not be Markov even when the augmented simulator state is. Comparing persistent and Markovized prompts tests the effect of retaining history; comparing genuine history with a scrambled, format-matched control helps separate temporally related information from the mere presence of additional prompt material. Temporal correlation then measures persistence, while path reversal tests ordering in the projected trajectory. State-space currents and trajectory reversal have exact thermodynamic interpretations in appropriate fully observed Markov systems [20, 21], but coarse-graining can hide dissipation and make inferred bounds estimator-dependent [22, 23]. The path statistic used here is therefore projected temporal asymmetry, not literal entropy production.

Recent LLM-population studies increasingly approach these questions with statistical-physics concepts: group size can alter collective misalignment [24], topology and self–social weighting can change conformity [25], intrinsic fields can outweigh fitted neighbor coupling [26], debate noise can produce model-dependent finite-size crossovers [27], and high-dimensional interaction data can admit compact dynamical regimes [28]. This progress narrows the gap, but the joint problem remains insufficiently resolved: controlled state and information boundaries, field reversal and restoration, genuine and placebo history interventions, spatial and temporal diagnostics, cluster-level inference, and an out-of-sample closure test are rarely examined together. A task score or single consensus coordinate cannot answer all of these questions.

The present design combines measurements because each resolves a different ambiguity. Belief and action order, overlap, effective compatibility, fluctuations, entropy, and dependence characterize alignment and synchronous organization. Connected graph-distance correlation detects spatial reorganization that mean order can miss; finite-window autocorrelation and Binder statistics describe persistence and order-parameter shape without establishing equilibrium correlation times, critical slowing down, or a phase transition. Coarse-grained block reversal compares temporal ordering across Markovized, persistent, and scrambled-history conditions. Matched quench controls and exact inference over independent graph–environment clusters define the evidential unit. Finally, an out-of-sample kinetic surrogate asks whether a small set of macroscopic coordinates can reproduce aspects of the full response and, equally importantly, where that reduced closure fails.

The contribution is not a general-purpose agent framework, a performance benchmark, or a model of human society. It is a controlled finite-size study that treats interacting LLM instances as measurable stochastic components and examines their collective response to external fields, retained and control histories, network organization, and coarse-graining. The direct experiments use two pinned instruction-model families, $N = 16$ reciprocal modular networks, and finite observation windows; the interpretation is effective and model-specific, with no thermodynamic-limit, exact physical entropy-production, or universality claim. Section 2 defines the network and interventions, Section 3 defines measurement and inference, and Section 4 sets out the study roles. Sections 5–6 present the quench and memory evidence, Section 7 tests reduced closure, and Sections 8–9 give the interpretation and conclusions.

2 State-separated LLM-agent networks and quench protocols

2.1 Network state and update dynamics

For agent i at attempted update t, the primary categorical coordinates are belief $b_i(t) \in \{-1, +1\}$ and committed action $a_i(t) \in \{-1, +1\}$. Its bounded local state also includes confidence, commitment status, workload, a memory code, at most three private history entries, an inbox, and an outbox. Display labels are counterbalanced and decoded back to latent spins. The microscopic projection is

$$Y_t = (\boldsymbol{b}_t, \boldsymbol{a}_t, \boldsymbol{c}_t, \boldsymbol{m}_t), \tag{1}$$

but this is not the complete simulator state.

Let

$$\Xi_t = (S_{1,t}, \ldots, S_{N,t}, G_t, \mathcal{D}_t, q_t, R_t) \tag{2}$$

denote all agent-private states $S_{i,t}$, graph state G_t, delivery operator $\mathcal{D}_t$, quench phase q_t, and the explicitly specified randomness state R_t. With fixed model weights and tokenizer, reconstructed local prompts, and a defined scheduler, Ξ_t induces a time-inhomogeneous transition kernel under the external quench protocol. The evaluator's map $Y_t = \phi(\Xi_t)$ deliberately omits some prompt history and randomness. Consequently Y_t and any rolling macrostate need not be Markov.

Figure 1 summarizes the information boundaries and the augmented-to-macroscopic map.

At each attempted update, a random permutation gives every agent exactly one opportunity per sweep. Only the scheduled instance receives its private observation, current local state, currently delivered inbox, and, in a memory condition, its own bounded history. It returns a validated typed object containing belief, action, confidence, commitment, memory state, outgoing signal, and tool action. One bounded repair may correct syntax; an invalid result causes no centrally selected scientific action. A serialized packet is sent on one graph edge. Fingerprints of nonrecipient peer-private states are compared around every transition.

The corresponding information-boundary and implementation checks are described in Appendix D.

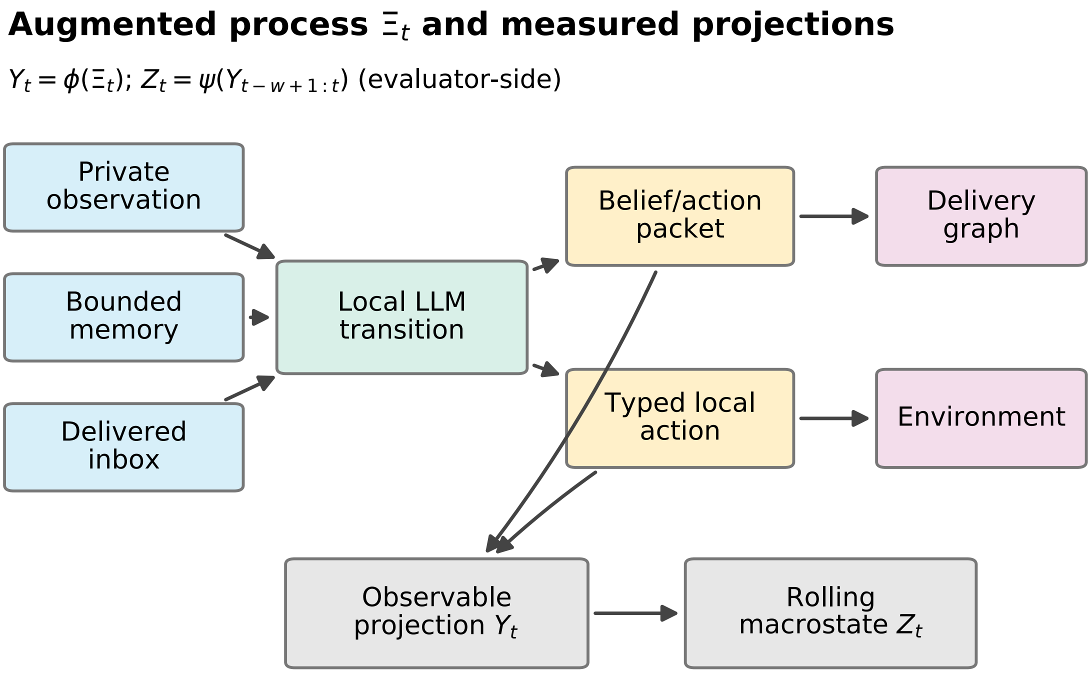

Figure 1: Agent information boundaries and the augmented-to-macroscopic map. The complete state Ξ_t includes private memory, graph and schedule state; the evaluator observes $Y_t = \phi(\Xi_t)$ and constructs rolling $Z_t = \psi(Y_{t-w+1:t})$. The scheduler offers a turn and delivers a packet but does not supply the model's belief, action, memory, or signal.

2.2 History conditions and matched controls

The Markovized prompt is reconstructed from the explicitly recorded current state and does not carry free conversational history. The persistent prompt adds the instance's last three own entries in the fixed format `time, belief, action, memory`. Thus two trajectories with identical visible $(\boldsymbol{b}_t, \boldsymbol{a}_t)$ can have distinct conditional future laws:

$$P(Y_{t+1} \mid Y_t, M_t) \neq P(Y_{t+1} \mid Y_t, M_t'). \tag{3}$$

If M_t is projected out, conditional dependence on older visible states can remain. This is a concrete mechanism for increased block reversal divergence, not a proof of physical dissipation.

The scrambled-history control isolates prompt presence and approximate length. It uses only the same agent's past opportunity times, while belief, action, and memory content are deterministically randomized from a prespecified control seed. No donor agent, future time, or hidden environmental outcome is used. Genuine and scrambled entries have the same section, count, field names, and format. Token counts and their paired differences are measured; turn-by-turn equality is not assumed.

Prompt-balance diagnostics for the scrambled-history control are shown in Figure 12 of Appendix B.

2.3 Field quench and restoration protocol

Each trajectory has 15 baseline sweeps, 15 intervention sweeps, and 15 restoration sweeps. In the field-quench arm the vector of balanced private fields changes sign at the first boundary and returns to its initial value at the second. Nominal trajectories retain the initial field. This follows the logic of a finite-system quench protocol, while making no quantum or thermodynamic-limit analogy. The initial change is the quench; restoration is the counter-quench.

Each family contributes six new graph/environment clusters. Each cluster has $N = 16$, a reciprocal modular fixed-degree graph, coupling $J = 0.8$, balanced disordered initialization, and four matched trajectories: nominal Markovized, field-quench Markovized, field-quench persistent memory, and field-quench scrambled history. Graphs, fields, initial states, label maps, update orders, recipient variates, and inference seeds are paired where technically meaningful.

3 Statistical-mechanical observables and inferential design

3.1 Macroscopic and information-theoretic observables

Belief and action order and their overlap are

$$m_b = \frac{1}{N}\sum_i b_i, \qquad m_a = \frac{1}{N}\sum_i a_i, \qquad q_{ba} = \frac{1}{N}\sum_i b_i a_i. \tag{4}$$

Disagreement is the fraction of connected pairs with unequal states. For a symmetric comparison layer $W_s = (W + W^\top)/2$, we define

$$H_{\text{ref}} = -\frac{J_b}{2}\boldsymbol{b}^\top W_s \boldsymbol{b} - \frac{J_a}{2}\boldsymbol{a}^\top D_s \boldsymbol{a} - K\boldsymbol{a}^\top \boldsymbol{b} - \boldsymbol{h}^\top \boldsymbol{b} - \boldsymbol{g}^\top \boldsymbol{a}. \tag{5}$$

We report $e_{\text{ref}} = H_{\text{ref}}/N$. Equation (5) is an effective compatibility coordinate anchored in a symmetric reference model. We do not infer a Gibbs law for the LLM process and do not call this literal energy. Likewise $N\,\text{Var}(e_{\text{ref}})$ is an energy-fluctuation observable, not heat capacity unless an independently justified effective temperature exists.

The belief susceptibility is reported without physical-temperature normalization,

$$\chi_b = N\,\text{Var}(m_b), \tag{6}$$

within a prescribed rolling window.

For discretized collective words x in a rolling window, Shannon configuration entropy is

$$S_{\text{config}} = -\sum_x \hat{p}(x)\log\hat{p}(x). \tag{7}$$

This is the usual plug-in Shannon functional [29]. The entropy rate is a bounded-history plug-in estimate of next-word unpredictability. Single-variable marginal entropy measures local occupancy; pairwise mutual information measures dependence between variables. Total correlation [30],

$$\mathcal{T} = \sum_{k=1}^{2N} H(X_k) - H(X_1, \ldots, X_{2N}), \tag{8}$$

separates marginal uncertainty from synchronous joint dependence.

The plug-in joint entropy of 32 binary variables in a short window has a large finite-sample floor [31]. We therefore retain the raw estimate but independently circular-shift each variable 200 times. Shifts preserve each marginal occupancy and within-variable temporal ordering while breaking synchronous cross-variable relations. We report raw, null mean, their untruncated difference, and a normalized difference divided by the sum of marginal entropies. The same audit is applied to pairwise and edge mutual information. A negative adjusted estimate remains negative.

The primary five-sweep representation is

$$\begin{aligned} Z_t = (&m_b, m_a, q_{ba}, e_{\mathrm{ref}}, v_E, S_{\mathrm{config}}, h_{\mathrm{rate}}, \mathcal{T}_{\mathrm{adj}}, I_{\mathrm{pair,adj}}, I_{\mathrm{edge,adj}}, \\ &\chi_b, \rho_b, D_b)_t. \end{aligned} \tag{9}$$

Three- and seven-sweep windows are prespecified sensitivities. This is an observable reduced representation, not a complete thermodynamic state.

3.2 Temporal asymmetry, spatial correlation, and persistence diagnostics

For a discrete observed word sequence and block length L, let $\hat{P}_L(x_0, \ldots, x_L)$ be the regularized empirical forward block law. We compute

$$\mathcal{I}_L = \frac{1}{L} D_{\mathrm{KL}} \left[\hat{P}_L(x_0, \ldots, x_L) \parallel \hat{P}_L(x_L, \ldots, x_0) \right] \tag{10}$$

in nats per attempted update. The primary $L = 3$ estimator uses pseudocount 0.5. Values at $L \in \{2, 4\}$ and pseudocounts 0.1 and 1.0 are sensitivities. Five hundred time permutations provide a panel-specific finite-length floor, and the adjusted statistic is the observed value in equation (10) minus its mean floor. Matched arms share the same shuffle tape. The machine-readable sensitivity table retains both the empirical permutation-null interval and the Monte Carlo standard error of its mean. These quantify different objects: dispersion of the null statistic and numerical uncertainty in the estimated bias floor, respectively. The added uncertainty columns reproduce, rather than replace, the prespecified floor mean and adjusted contrast.

For a fully observed stationary Markov chain with transition kernel P and stationary law π, the current

$$J_{xy} = \pi_x P_{xy} - \pi_y P_{yx} \tag{11}$$

can define an exact Markov entropy-production rate under the usual support conditions. We do not apply that interpretation to projected LLM trajectories. Persistent histories, semantic content, and bounded observations can violate first-order closure. Equation (10) is instead a coarse-grained time-reversal diagnostic and, under additional assumptions, may be a lower bound on hidden irreversibility [21, 23, 32].

Three secondary observables resolve spatial organization, temporal persistence, and the shape of the finite-size order-parameter distribution. First, for a phase window W we define the connected belief correlation at graph distance d,

$$C_b(d; W) = \frac{1}{|\mathcal{P}_d|} \sum_{(i,j) \in \mathcal{P}_d} \left[\langle b_i b_j \rangle_W - \langle b_i \rangle_W \langle b_j \rangle_W \right], \qquad \mathcal{P}_d = \{(i,j) : i < j,\ d_G(i,j) = d\}. \tag{12}$$

Shortest paths use the actual reciprocal modular delivery support. The bracket averages post-update observations at attempted-update resolution within one phase of one trajectory; node pairs are averaged inside that trajectory and never counted as independent replicates.

Second, the normalized magnetization autocorrelation and its fixed truncation are

$$\rho_m(\ell) = \frac{\langle [m_b(t) - \bar{m}_b]\,[m_b(t+\ell) - \bar{m}_b] \rangle}{\langle [m_b(t) - \bar{m}_b]^2 \rangle},$$
$$\tau_{\text{int}}^{(\ell_{\max})} = \frac{1}{2} + \sum_{\ell=1}^{\ell_{\max}} \rho_m(\ell). \tag{13}$$

The primary descriptive truncation is $\ell_{\max} = 2N = 32$ attempted updates; N and $3N$ are fixed sensitivities. Zero-variance phase series are left undefined rather than assigned artificial persistence. The quantity is a finite-window correlation sum reported in attempted-update units, not evidence of critical slowing down or an effective sample-size estimate. Its absolute short-lag scale includes persistence induced by changing at most one agent on each random-sequential update; matched arms share that update convention. Quench and restoration windows can also be nonstationary, so their sums are finite-window persistence diagnostics rather than equilibrium integrated correlation times.

Third, we calculate the Binder cumulant

$$U_4 = 1 - \frac{\langle m_b^4 \rangle}{3 \langle m_b^2 \rangle^2} \tag{14}$$

separately for each phase and trajectory [33]. Near-zero second moments are reported as undefined. With only $N = 16$, equation (14) is a finite-size distribution-shape diagnostic; no crossing, critical point, or thermodynamic-limit transition is inferred. A post-reconstruction estimator sensitivity repeats the calculation over each full phase and its early and late halves. It compares the mean of the six trajectory-level cumulants with a pooled-moment construction; both uncertainty calculations resample complete graph–environment clusters, never updates.

3.3 Recovery estimands and exact inference

For each held-out graph/environment cluster and model, the nominal manifold is fitted only from nominal trajectories in the other five clusters. Training medians impute nonfinite coordinates; training means and scales standardize features; shrinkage covariance with a prespecified ridge fraction defines a Mahalanobis-like distance. The nominal 95th-percentile threshold is computed from those training distances. The held-out cluster contributes neither a scale nor a threshold.

We summarize peak departure, integrated response, path length, final residual, and threshold re-entry. The prospective cross-model recovery contrast is

$$\Delta d_{\text{rec}} = \bar{d}_{31:35} - \bar{d}_{41:45}, \tag{15}$$

which can be positive, zero, or negative. It replaces a historical maximum minus final-window quantity whose sign was structurally constrained.

A complete graph/environment trajectory cluster is the inferential unit.

The hypotheses were specified before the cross-model outcomes were examined. H1 compares Granite field-quench peak departure with matched nominal evolution at one-sided $\alpha = 0.02$. H2 compares persistent with Markovized adjusted reversal divergence. H3 compares persistent with scrambled history. H4 tests the fixed recovery contrast. H2–H4 form a one-sided Holm family with total $\alpha = 0.03$. Exact cluster sign flips are the primary tests; deterministic 10,000 cluster bootstraps give intervals. The sign-flip enumeration is exact conditional on the observed paired magnitudes under a sign-symmetry null; it is not described as treatment- label randomization. Agent and time-point counts do not increase the inferential sample size.

The exact-test, multiplicity, and cluster-bootstrap details are retained in Appendix A.

4 Study design and evidential roles

4.1 Exploratory discovery and corrected earlier evidence

The memory evidence developed in three stages: an exploratory experiment, a subsequent prospective replication, and the present cross-model extension with an additional history control. Figure 2 displays these stages separately so that the earlier estimates are not retroactively pooled into the cross-model confirmatory interval.

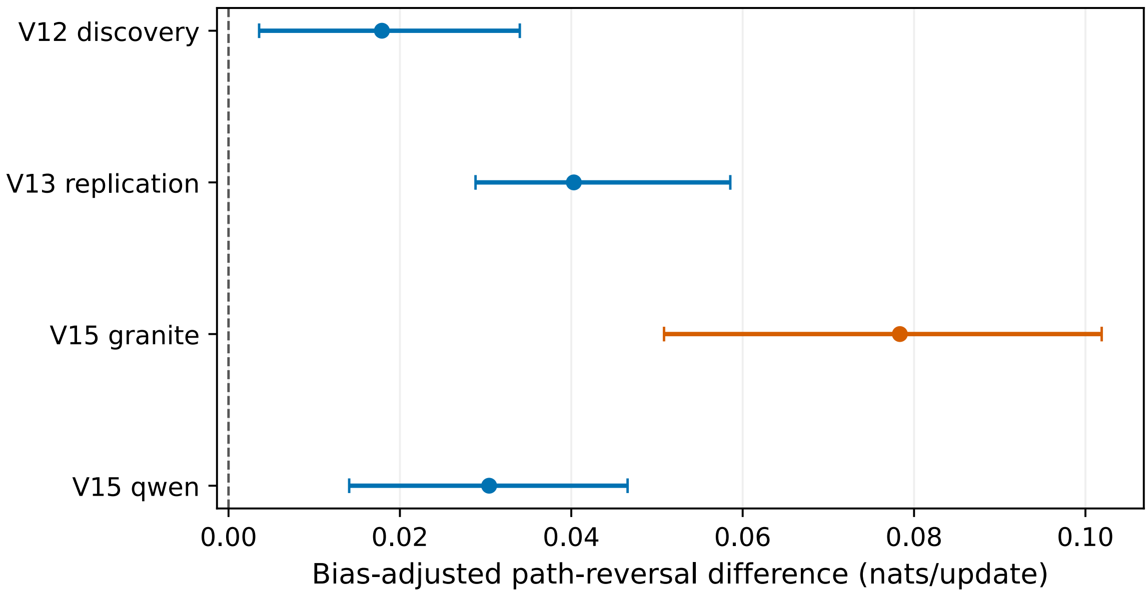

Figure 2: Exploratory discovery, prospective replication, and cross-model memory contrasts are shown separately. Points are graph/environment-cluster effects; intervals resample complete clusters. The exploratory and replication-stage estimates are not retroactively pooled into the cross-model confirmatory interval.

The audit of the earlier quench experiment preserves every recorded prompt, completion, categorical choice, and trajectory. The original analysis computed a field panel's recovery threshold from its own baseline, despite protocol text specifying a training-only threshold. The corrected analysis passes an explicit cluster-to-training-threshold map from leave-one-cluster-out fitting into the recovery summaries.

The earlier study's third hypothesis used the maximum recovery distance minus the final-five mean. That value remains numerically reproducible, but the construction is nonnegative for nearly every nonconstant trajectory. Its raw and Holm-adjusted sign-flip probabilities are retained as historical audit values and removed from inferential support. The corrected report distinguishes whether the numerical criterion was met, whether a directional test is valid, and whether the time-resolved path is consistent with return.

Figure 3 displays the corrected time series from the earlier quench experiment.

Cluster-level recovery and delayed-audit material is retained in Appendix A, including Figures 9 and 10.

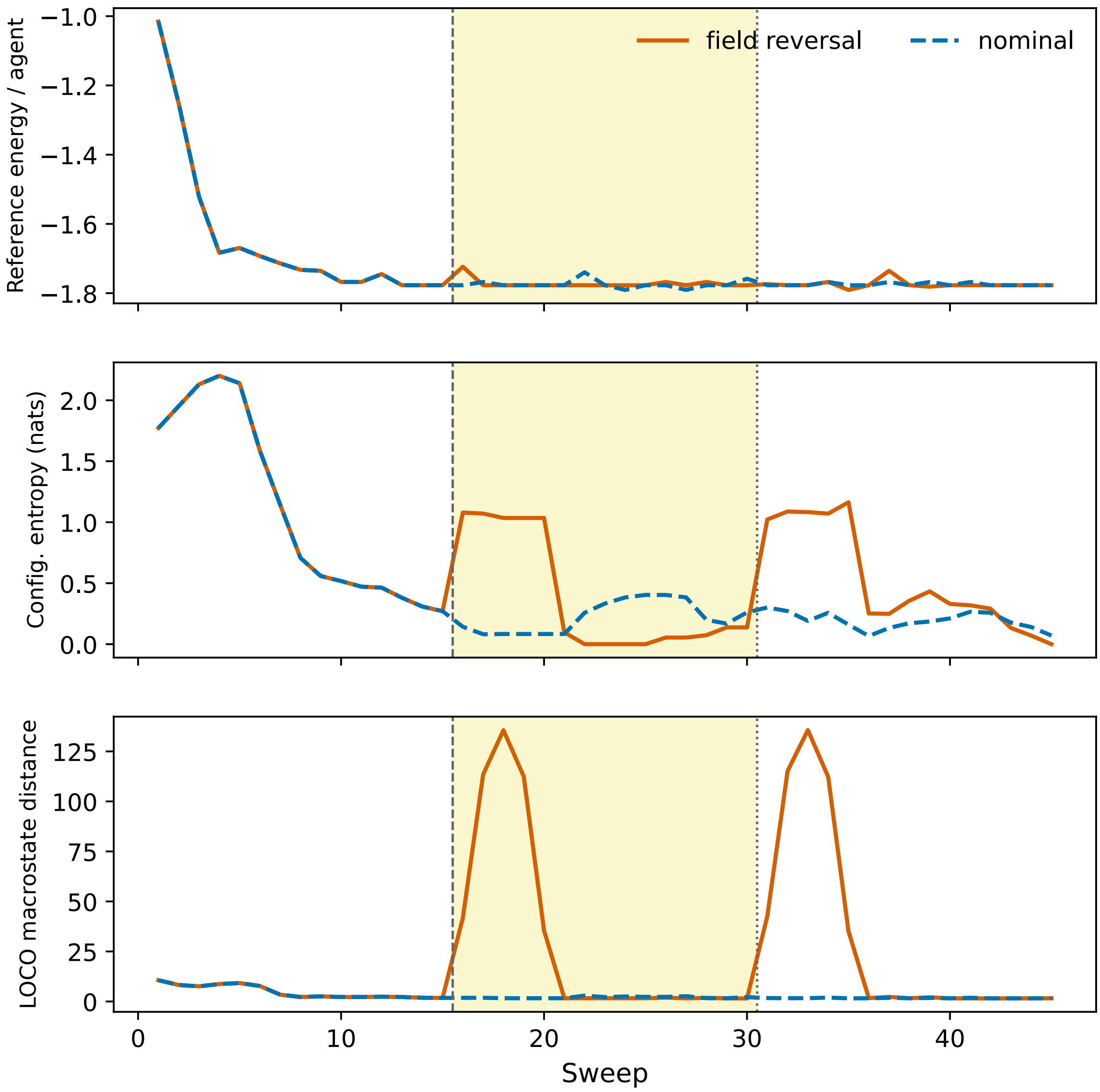

Figure 3: Corrected time series from the earlier experiment for nominal and field-reversal trajectories. The yellow interval marks the quench; the dotted boundary marks restoration. Reference energy is effective, entropy is configuration diversity, and distance is measured in the cluster-excluded nominal geometry.

4.2 Cross-model confirmation and control logic

The original model family is represented by Qwen/Qwen2.5-7B-Instruct at immutable revision `a09a35458c702b33eeacc393d103063234e8bc28`. The independent family is IBM Granite-3.3-8B-Instruct at revision `51dd4bc2ade4059a6bd87649d68aa11e4fb2529b` [34, 35]. Both use the same pinned inference stack, NF4 double quantization, BF16 computation, decoding temperature 0.5, top-$p = 0.9$, and at most 96 generated tokens. The decoding temperature is a model-generation setting, not an inferred thermodynamic temperature.

The pre-outcome engineering pilot was restricted to loading, schema validity, occupancy, both transition directions, timing, information isolation, delivery, prompt-token balance, latency, and projected compute. It does not calculate a quench, memory, or irreversibility contrast. The prospective design contains 48 trajectories and 34,560 attempted local decisions. Complete unfavorable trajectories are retained.

5 Quench response, restoration, and finite-size organization

5.1 Cross-model response and recovery

The Granite-family H1 contrast is 42.263 distance units (95% interval 24.316 to 59.303); its prespecified disposition is supported. Figure 4 displays model-specific distances rather than pooling their absolute scales. A positive contrast indicates a field-driven departure beyond nominal variation; a nonpositive or uncertain contrast would delimit cross-family replication. The complete cluster table is retained regardless of direction.

All six Granite cluster effects are positive. The separately retained Qwen contrast is also positive in all six clusters, with a mean field-minus-nominal peak of 112.691 units. These absolute values are not compared across models: each leave-one-cluster-out geometry is standardized and fitted within model.

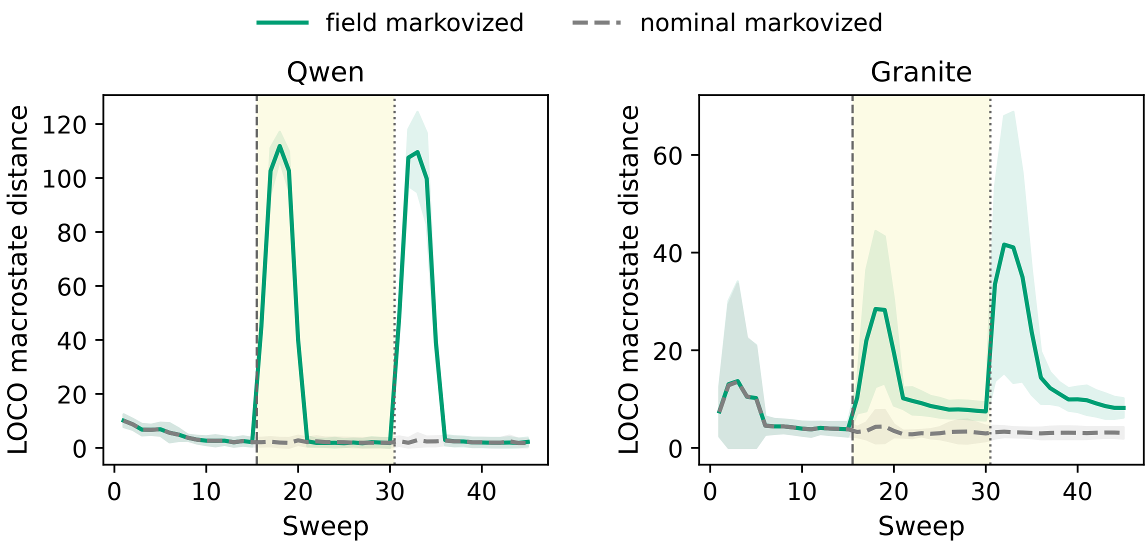

Figure 4: Model-specific field-quench and nominal trajectories. Curves are means over six graph/environment clusters and bands show between-cluster dispersion. Nominal fits, scaling, covariance, and thresholds exclude the displayed cluster.

Restoration H4 is 52.541 distance units (95% interval 35.406 to 68.673), with prespecified disposition supported. Unlike the earlier experiment's historical recovery statistic, this fixed-window quantity permits either sign. Recovery time and threshold crossing remain trajectory descriptors, not an equilibrium relaxation theorem. The quench and counter-quench paths need not retrace one another because messages, actions, and history alter the state reached at restoration.

The fixed-window decline occurs in every model–cluster pair, but it does not imply complete return within the observation horizon. Every Qwen trajectory re-enters its training-nominal threshold six sweeps after restoration. Only one of six Granite trajectories re-enters by sweep 45; the other five retain final-five means above their own thresholds. Thus H4 supports motion toward the restored nominal regime, while Granite exhibits slower or incomplete finite-horizon recovery.

5.2 Spatial and finite-window organization

Equations (12)–(14) are secondary descriptive analyses of the same recorded trajectories; they neither modify the confirmatory estimands nor introduce additional simulation arms. They were specified during reconstruction after the confirmatory aggregate results were known and are therefore not a new prospective hypothesis family. Connected correlation matrices are first calculated within every complete model–graph–environment trajectory and phase. Their graph-distance profiles are then summarized over the six independent clusters. Figure 5 shows cluster-averaged matrices for the persistent- history quench window and phase-resolved distance profiles. Community boundaries are fixed by the graph construction rather than selected from the observed correlations.

At graph distance one, the persistent-history quench-minus-baseline connected correlation changes are 0.00238 (-0.01750 to 0.02277) for Qwen and -0.02032 (-0.05486 to 0.00947) for Granite. These cluster-level descriptive intervals quantify finite-window reorganization; their signs are not assigned a confirmatory disposition.

Appendix C and figure 14 combine the full autocorrelation curve with the fixed-lag integral in equation (13), the Binder statistic in equation (14), and empirical magnetization occupancy. This combination prevents a Binder value from being interpreted without its underlying finite-window distribution. It also separates persistence of the projected collective coordinate from the block-reversal statistic: the former is time-symmetric two-point dependence, whereas the latter compares forward and reversed path words. The accompanying source table also retains full-, early-half-, and late-half Binder estimates under cluster-first and pooled-moment rules. Their differences diagnose finite-window and aggregation sensitivity rather than selecting whichever construction yields the larger separation.

At the fixed two-sweep truncation, persistent-minus-Markovized recovery τ_{int} changes are -2.46995 (-5.79214 to 0.17724) updates for Qwen and -4.14109 (-19.88286 to 11.60067) updates for Granite. The field-Markovized quench-minus-baseline Binder changes are -4.80504 (-7.42824 to -2.20832) and 0.16643 (-0.09447 to 0.52063), respectively. These estimates are reported whether positive, negative, or interval-compatible with zero.

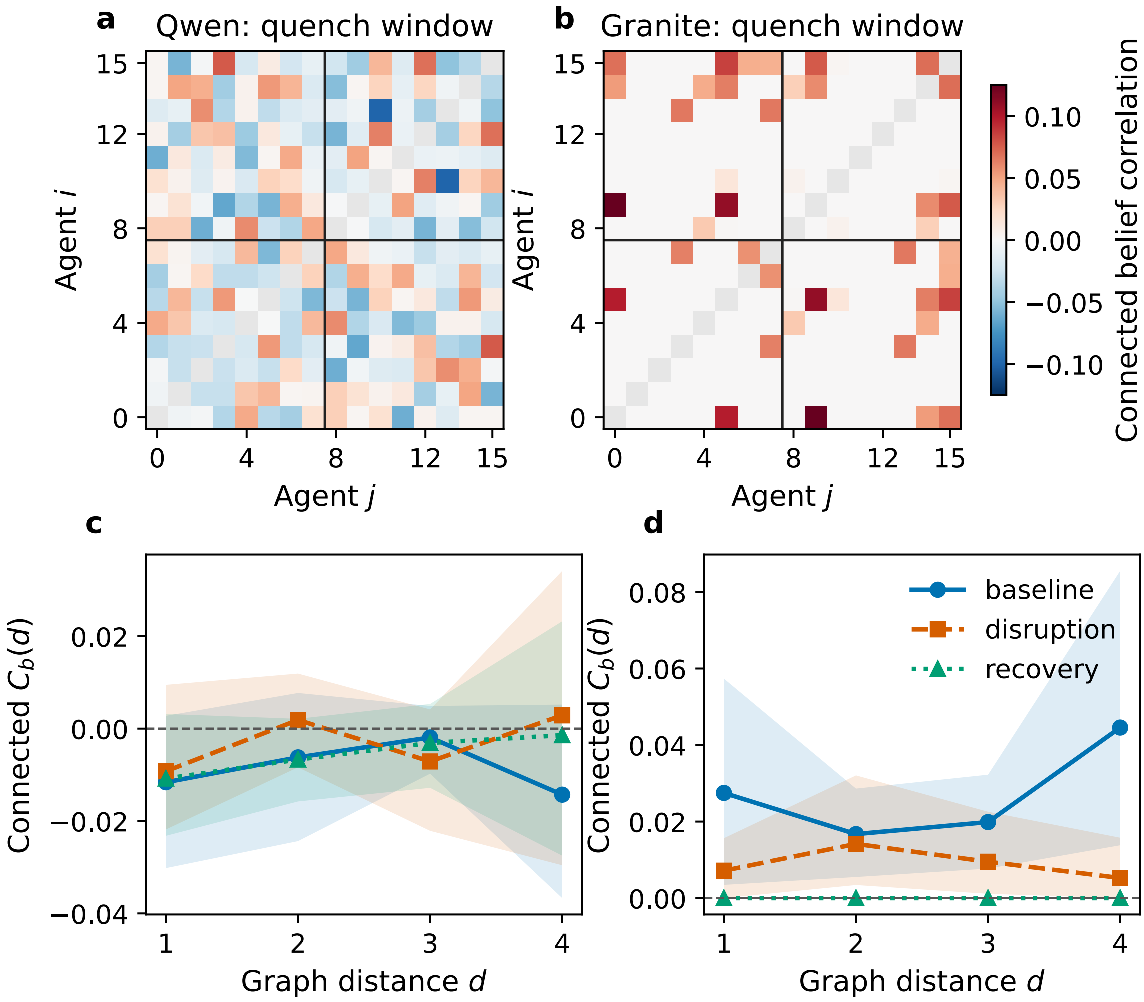

Figure 5: Connected belief correlations in the persistent-history arm. The upper panels average complete 16×16 connected-correlation matrices over all six clusters in each model; black rules mark the two predefined communities. Agent indices align across graph realizations only through those community memberships, so fine node-level texture in an averaged matrix has no cross-realization identity interpretation. The lower panels show equation (12) by graph distance, with 95% complete-cluster bootstrap intervals. Individual pairs and updates do not enter the inferential sample size.

6 Memory and projected temporal asymmetry

6.1 Persistent, Markovized, and scrambled histories

Relative to Markovized state, genuine memory changes adjusted block divergence by 0.05438 nats per attempted update (95% interval 0.03459 to 0.07548); the prespecified disposition is supported. H2 asks whether history matters relative to an explicit current-state prompt.

Relative to the scrambled-history placebo, the change is 0.04845 (0.02190 to 0.07526), disposition supported. H3 is more stringent: it asks whether the temporally related content matters beyond adding a similarly shaped prompt section.

Several arm-level adjusted values are negative because their observed block divergence lies below the corresponding finite-sample shuffle-floor mean. H2 and H3 are paired differences between these untruncated adjusted statistics; their positive contrasts do not establish positive absolute entropy production in either arm.

No exact thermodynamic interpretation follows from a positive result. The observed block words omit natural-language content and some private state. Conversely, a null H3 would be informative: prompt presence, bounded memory depth, model-specific history use, or estimator power could explain the earlier exploratory and replication-stage contrast without a reproducible content-specific effect. Block-length, pseudocount, conditional-memory-depth, occupancy, and token-balance tables support this distinction.

Figure 6 displays the paired cluster contrasts for the two pinned model families.

The prompt-balance diagnostic in figure 12 of Appendix B records the scope of the format and approximate-length control.

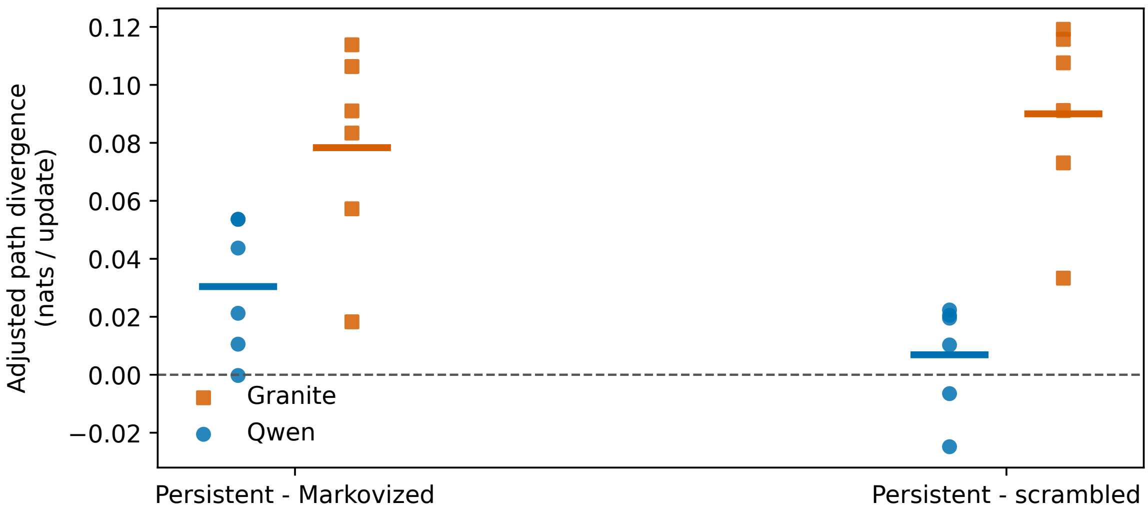

Figure 6: Per-cluster genuine-memory contrasts for both pinned model families. Short horizontal segments indicate model means. The scrambled-history arm controls prompt section and approximate length while destroying temporal relation to the current trajectory.

6.2 Model heterogeneity and estimator dependence

The prospectively pooled contrast retains substantial model heterogeneity. For persistent minus Markovized state, model-specific means are 0.03041 for Qwen (five of six cluster effects positive) and 0.07835 for Granite (six of six positive). For persistent minus scrambled history they are 0.00689 for Qwen (four of six positive) and 0.09001 for Granite (six of six positive). The model-specific Qwen scrambled-control sign-flip value is 0.21875 and is not separately confirmatory; the prespecified pooled H3 remains supported because model-stratified cluster pairs were the prespecified units. This decomposition supports cross-family direction for the Markovized contrast but not a claim that content-specific memory effects are uniform across model families.

Estimator sensitivity further limits the scale claim. The pooled memory contrasts remain positive over most prespecified block-length and pseudocount combinations, but their magnitude varies substantially; at block length four and pseudocount one, persistent-minus-scrambled is only 0.00024 nats per update and the Granite component is slightly negative. Conditional-memory information is likewise not monotonically larger in the persistent arm. Accordingly, the robust statement concerns the prespecified coarse graining, its bias floor, and paired controls rather than an estimator-independent entropy- production rate.

The block-length, pseudocount, and shuffled-floor sensitivities are displayed in figure 11 of Appendix B.

Taken together, H1–H4 address distinct claims rather than a single composite effect: H1 tests field-driven departure, H2 tests persistent history relative to current-state prompting, H3 tests temporally related content relative to a format-matched control, and H4 tests fixed-window restoration. Figure 7 places the four prespecified effects and their cluster-bootstrap intervals in one synthesis while keeping their units separate. Under the prespecified tests and multiplicity allocation, all four criteria are met; the model-specific and estimator boundaries above remain part of that interpretation.

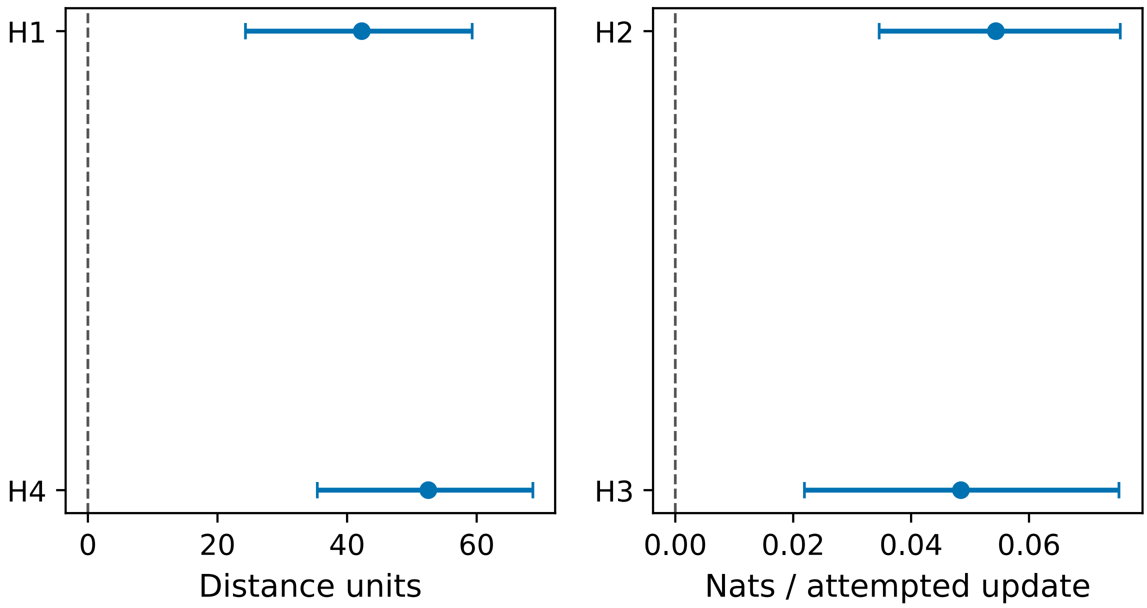

Figure 7: Prespecified effects and 95% cluster-bootstrap intervals. Distance and information units occupy separate panels; no claims forest places incompatible quantities on one numerical axis. Dispositions use exact cluster sign flips and the prespecified multiplicity allocation.

7 Reduced dynamical descriptions and closure limits

7.1 Information retained beyond mean order and the kinetic surrogate

Magnetization records directional order but can remain near zero when half the population reorganizes in opposing communities. Configuration entropy then records collective state diversity; marginal entropy records local occupancy; total correlation distinguishes independent uncertainty from coordinated dependence; effective energy records compatibility with the symmetric reference; susceptibility and energy variance record fluctuations; autocorrelation records finite-window persistence; and path reversal records temporal ordering. These quantities are complementary only when their estimators and scales are kept distinct.

The representation analysis of the earlier quench experiment uses transparent multinomial logistic regression with leave-one-cluster-out preprocessing. The delayed permutation audit tests both absolute balanced accuracy and the increment over order-only features. This is not a claim that entropy outperforms all ordinary summaries. It tests whether a predefined collective representation retains condition information discarded by a reduced representation. The field-quench trajectory supplies a mechanistic example: energy, dependence, entropy rate, and fluctuation can change while mean magnetization remains balanced.

We fit logistic belief and action responses only to the immutable microscopic-response table from the prospective replication stage. The belief model includes private field, delivered neighbor field times controlled coupling, previous belief, and previous action; the action model includes updated belief and previous action. No coefficient is refitted to either later quench trajectory set. The surrogate is run on the same graph generator, initial-state generator, asynchronous update tape, field schedule, and intervention times. It is a deliberately low-dimensional kinetic-Ising comparison rather than a fitted description of every asymmetric or history-dependent response [36].

Figure 8 compares the direct trajectories from the earlier quench experiment with the out-of-sample kinetic surrogate.

In the five-coordinate comparison geometry, the direct field-minus-nominal peak-response difference is only 0.142 units on average and is nonzero in one of six clusters, whereas the surrogate difference is 1.467 units and is positive in four clusters. The corresponding field-minus-nominal energy–entropy route-area differences are 0.043 and 0.695. The direct shared response peaks at the quench boundary in one cluster and is otherwise dominated by initial relaxation; the surrogate places its peak at the quench or counter-quench boundary in four clusters. Both descriptions return close to their own final shared-response baseline, but the surrogate neither reproduces the direct amplitude nor its timing. This also explains why the richer macrostate distance used in the earlier quench analysis can show a large quench pulse that a short kinetic closure over a small shared coordinate set misses.

7.2 Captured response, failure modes, and finite-size context

This comparison identifies three possible failure modes. First, a homogeneous logistic neighbor coefficient cannot capture state-dependent semantic use. Second, action and belief persistence can introduce lags not represented by a single synchronous mean field. Third, hidden memory and workload can produce different future laws at the same projected spin configuration. Adding unconstrained parameters after observing quench trajectories would obscure these limits, so the closure remains low-dimensional and out of sample.

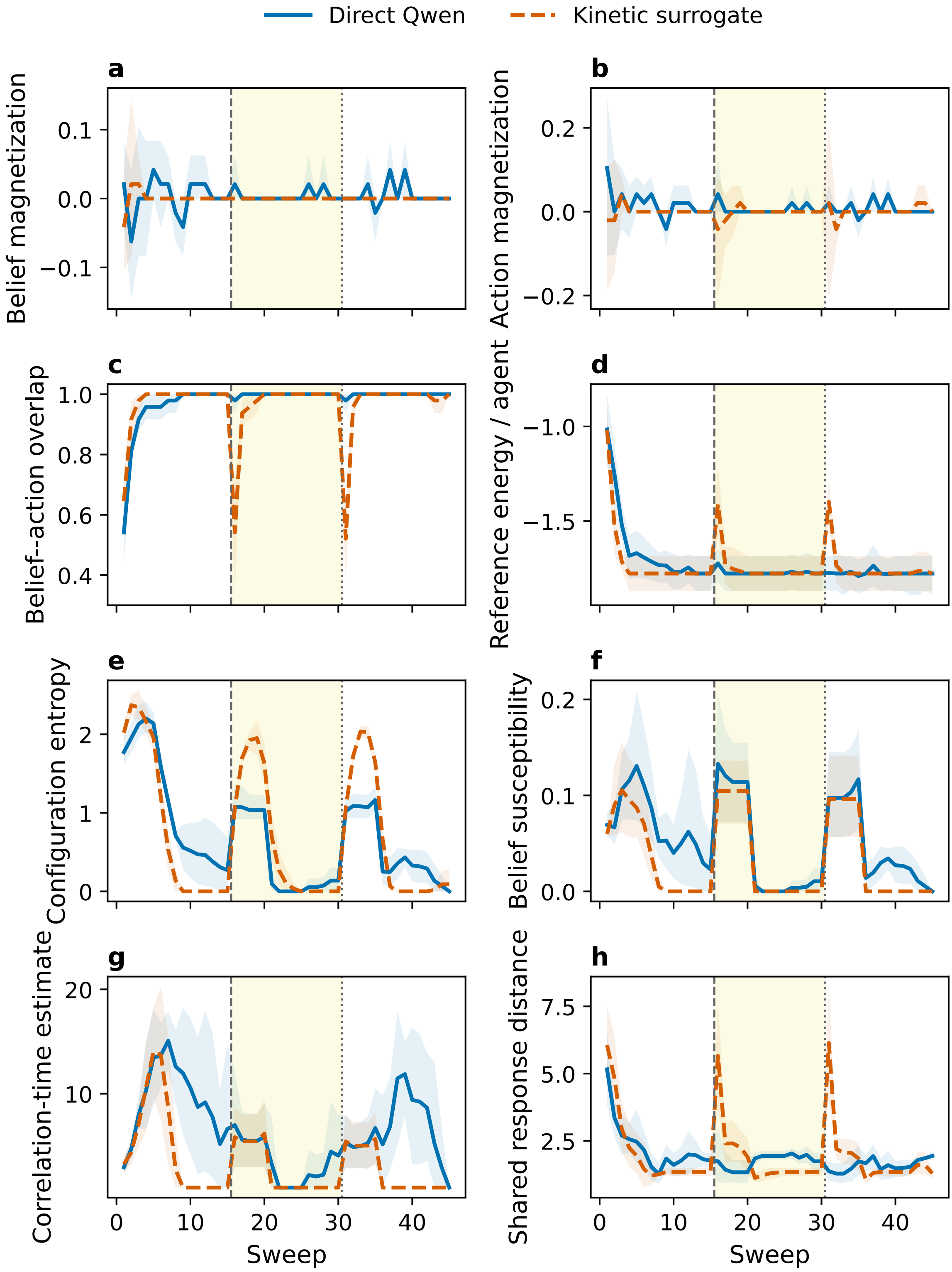

Figure 8: Out-of-sample field-quench comparison between direct Qwen trajectories and the replication-stage kinetic surrogate. Matching direction, peak timing, normalized amplitude, recovery, and route area are assessed separately. Surrogate failure is evidence about the limits of the closure, not a failed LLM experiment.

CPU trajectories at $N \in \{8, 16, 32, 64\}$ supply a dense effective-model size context. They are not direct LLM finite-size scaling and cannot establish a critical exponent or thermodynamic-limit phase transition. The direct $N = 16$ paths remain empirical anchors.

The effective-model size context is displayed in figure 13 of Appendix B.

8 Discussion

Treating the LLM-agent network as a stochastic interacting system begins with implemented rather than metaphorical objects: agent instances receive local prompts, exchange packets, update on a recorded schedule, and change categorical state. Magnetization, Shannon entropy, mutual information, total correlation, and path divergence are then mathematical observables of those records. State separation is central to this interpretation because it distinguishes the complete augmented state that determines a transition from the local information available to an agent and from the lower-dimensional projection available for population-level measurement.

The quench experiments use this formulation to measure finite-horizon response and recovery rather than equilibrium relaxation. Both model families depart from their within-model nominal geometries after field reversal, and the fixed-window restoration contrast supports motion back toward the restored nominal regime. The difference between complete Qwen threshold re-entry and slower or incomplete Granite re-entry within the observation horizon remains important: the common directional response does not imply a common recovery time or complete return. Recovery times and threshold crossings are trajectory descriptors, and the quench and counter-quench paths need not retrace because messages, actions, and history change the state reached at restoration.

The history interventions clarify what memory means under coarse-grained observation. Persistent history differs from an explicit current-state prompt, and comparison with scrambled history asks whether temporally related content matters beyond prompt format and approximate length. The distinction between complete augmented state, observable projection, and rolling macrostate explains how private memory can generate history dependence in the projected process. Positive paired contrasts nevertheless do not establish positive absolute entropy production: some arm-level bias-adjusted values are negative, the observed path words omit natural-language content and private state, and the measured reversal divergence is a coarse-grained projected statistic rather than an exact physical entropy-production rate.

Model heterogeneity and negative findings limit generalization. The persistent-versus-Markovized contrast has the same direction across the two families, whereas the content-specific scrambled-history contrast is much weaker and less uniform for Qwen than for Granite. Estimator choices also change the magnitude of the memory contrasts. In the exploratory evidence, degree- and traffic-matched nonreciprocity did not produce a reliable increase in irreversibility, and partition and message-corruption trajectories in the earlier quench experiment remained close to nominal under their tested construction. These results prevent the broader assertion that any directed graph, entropy statistic, or history prompt produces the same collective effect.

The secondary spatial, persistence, Binder, and closure analyses add distinct but limited information. Connected correlations describe how finite-window organization changes across the fixed modular graph, while autocorrelation summarizes persistence of the projected collective coordinate. Binder values are shown with occupancy and aggregation sensitivities; they do not establish a crossing, critical slowing down, or a phase transition. The out-of-sample kinetic surrogate captures some direction and recovery features but misses parts of the direct response amplitude and timing. Its inexpensive size context is not direct-LLM finite-size scaling, so surrogate failure identifies limits of the reduced closure rather than a failed agent experiment.

These measurements support an effective statistical-mechanical description, not a literal identification of the network with a thermodynamic material. The reference energy is effective because the LLM transition rule is not derived from its Hamiltonian, and decoding temperature is a generation setting rather than an independently inferred thermodynamic temperature. No free-energy-like coordinate is emphasized because such a temperature is unavailable. The value of the framework is instead that it keeps collective organization, fluctuations, dependence, response, and temporal ordering operationally distinct while exposing the assumptions behind each reduction.

Several limitations bound the scope. Two 7B instruction-model families do not establish universality, and the direct systems contain only $N = 16$ agents, so no thermodynamic-limit transition is claimed. The binary projection compresses semantic messages and private state. The confirmatory study uses one reciprocal modular topology and one coupling/noise anchor. The scrambled-history placebo matches format and approximate token distribution but cannot make semantic content exchangeable turn by turn. Short-window entropy, dependence, autocorrelation, and path estimates remain sensitive to coarse graining and finite sampling even after their null-floor and aggregation checks.

Further work therefore requires larger direct systems, additional network topologies, more model families, and longer stationary trajectories. Richer observable-state representations could test which hidden variables account for projected memory, while prospectively chosen alternative coarse grainings could establish whether the present estimator dependence persists. Such work is needed before thermodynamic-limit, critical, or exact stochastic-thermodynamic claims would be warranted.

9 Conclusions

State-separated LLM-agent instances form a measurable interacting stochastic system whose collective trajectories admit a disciplined reduced description. Controlled field quench and restoration create a common test of order, uncertainty, dependence, effective compatibility, fluctuation, and temporal response. The prospective cross-model experiment adds an independent model family and a format-matched history control to the exploratory and earlier quench evidence. The pooled prospective memory effects are positive, while the weaker Qwen scrambled-control decomposition, Granite's incomplete threshold re-entry, and estimator sensitivity remain explicit boundary results.

The statistical-mechanical contribution is therefore not that entropy makes agents perform better. It is a formulation and measurement framework that connects local language-conditioned transitions to collective finite-size dynamics, exposes what population averages discard, and makes failures of simple closures explicit.

Data and code availability

Code, configurations, figure source data, aggregate results, and reproduction scripts are available at `https://github.com/mantzaris/ThermoAgent`. The reported cross-model experiment ran on an NVIDIA GeForce RTX 4090 and used approximately 19.19 measured generation GPU-hours. Raw prompts, completions, and full trajectories are not distributed because of size and potential sensitivity, and the original external raw tree is no longer available; content-addressed manifests and replay and aggregate-validation artifacts document the reconstruction.

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A Study chronology, correction, and inferential details

The earlier quench analysis originally derived each field panel's recovery threshold from that panel's own baseline, although the protocol called for a threshold fitted only to the training clusters. The corrected analysis maps the leave-one-cluster-out training threshold into the held-out recovery summary without changing any recorded trajectory. Figure 9 retains every cluster-level recovery path, including the high-response cluster, under the corrected training-only construction. The historical maximum-minus-final recovery statistic remains numerically reproducible but is not used as directional evidence because its construction is nonnegative for nearly every nonconstant trajectory.

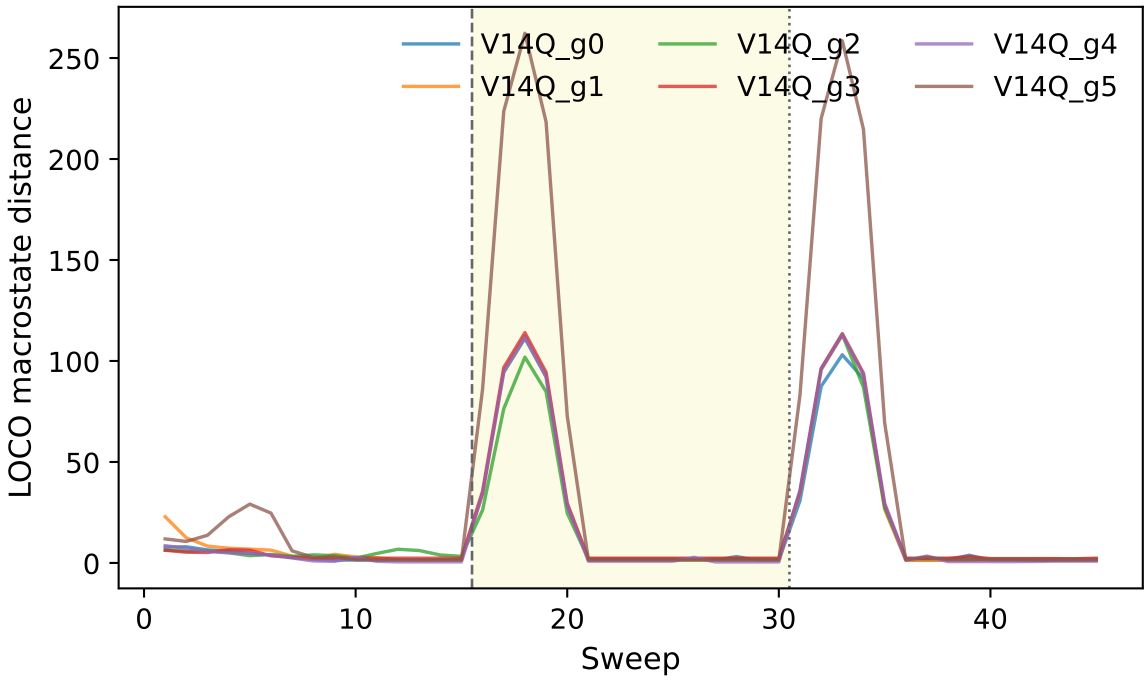

Figure 9: All field-reversal cluster trajectories from the earlier experiment, including the high-response cluster. The corrected audit replaces the held-out-panel threshold with the training-cluster nominal threshold. The historical recovery sign statistic is not used as directional evidence.

The delayed audit executes the omitted 10,000 cluster-preserving label permutations. All four panels stay together within each cluster, and every replicate refits imputation, scaling, logistic regression, and leave-one- cluster-out predictions. Three-, five-, and seven-sweep nominal geometries are fitted independently. Each coordinate and observable family is deleted in turn, and dependence estimates receive circular-shift null floors. These were delayed prespecified sensitivity analyses completed after the primary outcomes because they had been omitted from the initial implementation, not a new prospective experiment.

The representation permutation retains the four condition panels in every cluster and permutes labels only within cluster. For each of 10,000 replicates, every leave-one-cluster-out fold refits imputation, scaling, and the multinomial logistic model; test-cluster features do not enter preprocessing. Figure 10 records the delayed dependence and full-pipeline permutation audit.

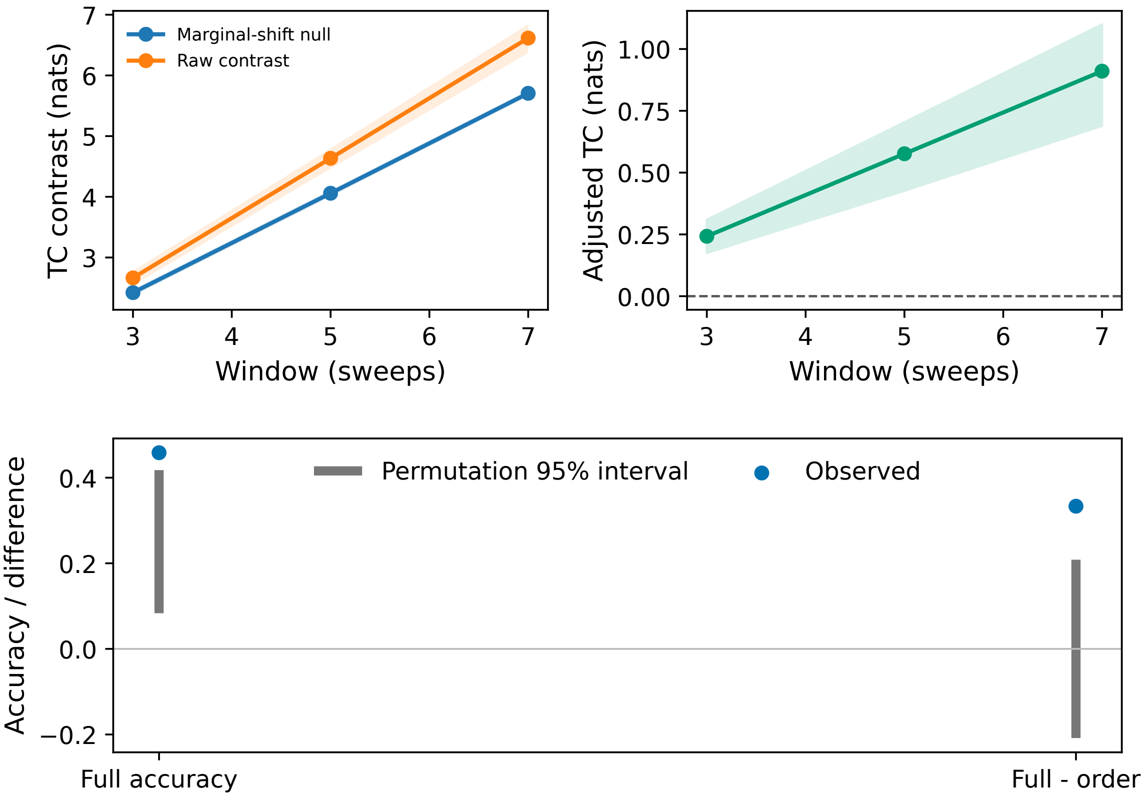

Figure 10: Delayed prespecified dependence and permutation audits. Raw and bias-adjusted dependence, window sensitivity, and cluster-preserving nulls are shown explicitly. These audits were completed after the primary outcomes because of implementation omissions.

For the prospective cross-model tests, the one-sided sign-flip probability enumerates all 2^n sign assignments of the observed cluster effects for $n \leq 20$ paired clusters. The cluster bootstrap samples complete graph/environment clusters with replacement and never resamples time points as independent observations. H1 receives alpha 0.02; H2–H4 receive a Holm-adjusted family alpha 0.03. The allocation was fixed before outcomes were examined. Intervals are uncertainty summaries and do not replace the exact paired tests.

B Estimator, control, and closure sensitivities

Total-correlation nulls independently circular-shift each coordinate by a nonzero offset where possible. This retains marginal occupancy and each coordinate's internal temporal ordering while breaking synchronous dependence. The adjusted estimate is not truncated. Alternative rolling windows rebuild the words, covariance, training threshold, and held-out distance rather than subsampling a five-sweep output.

For block reversal, time-shuffled floors quantify finite-length occupancy bias; they do not prove stationarity. Results at block lengths two, three, and four expose finite-history sensitivity, while conditional mutual information at history depths one through three provides a separate diagnostic of incomplete first-order closure. Figure 11 displays the prespecified estimator-sensitivity grid.

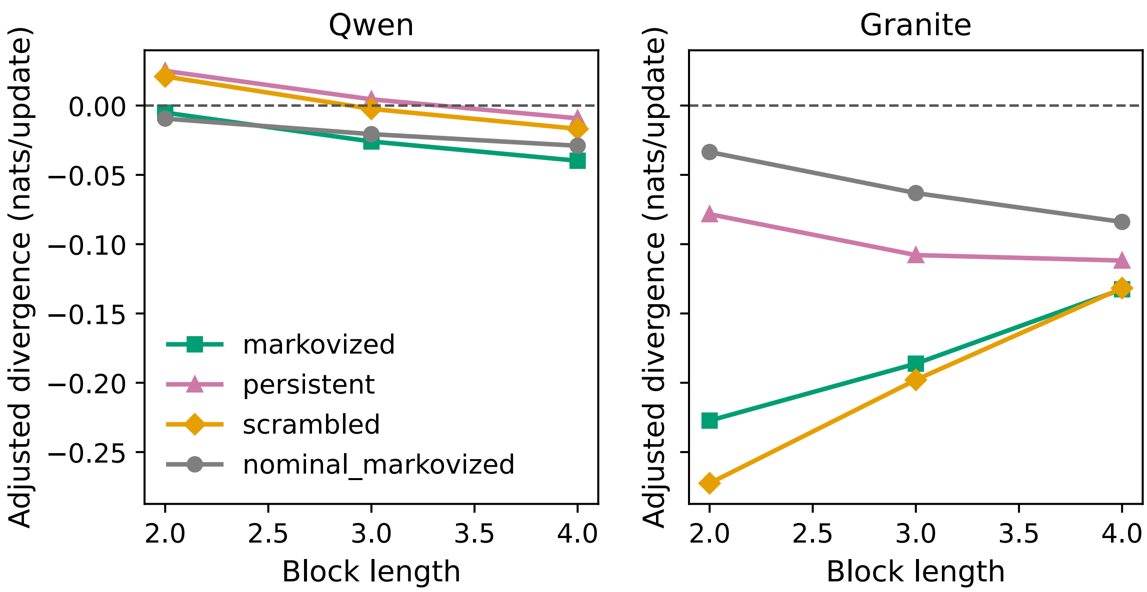

Figure 11: Block-length, pseudocount, and shuffled-bias-floor sensitivity for bias-adjusted block reversal divergence. Memory contrasts can be checked against estimator choices; observable coarse-graining and finite length remain limiting.

The scrambled-history arm controls the presence and approximate length of the history prompt while destroying its temporal relation to the current trajectory. Figure 12 records the token-count diagnostic for that comparison. It supports approximate prompt-length and format balance but does not establish turn-by-turn token identity or semantic exchangeability.

Finally, CPU trajectories at several sizes provide context for the reduced kinetic model. Figure 13 places the direct-system anchor beside that effective-model size comparison. These calculations are not direct-LLM finite-size scaling and do not support a thermodynamic-limit claim.

C Finite-window persistence and Binder diagnostics

The autocorrelation and Binder diagnostics defined in equations (13) and (14) are summarized in figure 14. Each quantity is computed within one complete model–graph–environment trajectory before uncertainty is summarized across the six clusters per model. The autocorrelation sum uses the fixed two-sweep truncation, while one- and three-sweep truncations remain machine-readable sensitivity analyses. Zero-variance phase series are left undefined rather than assigned artificial persistence, and uncertainty resamples complete clusters rather than treating time points as independent.

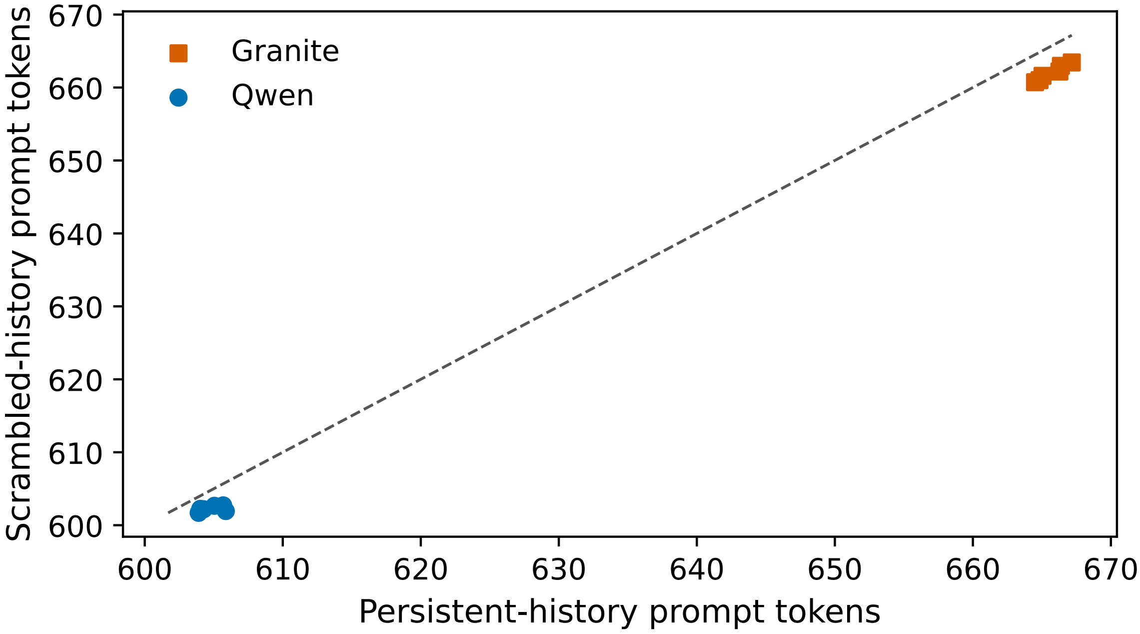

Figure 12: Scrambled-history prompt-length control. Per-cluster mean token counts verify that prompt length and format are approximately matched; semantic content cannot be exactly token-matched turn by turn.

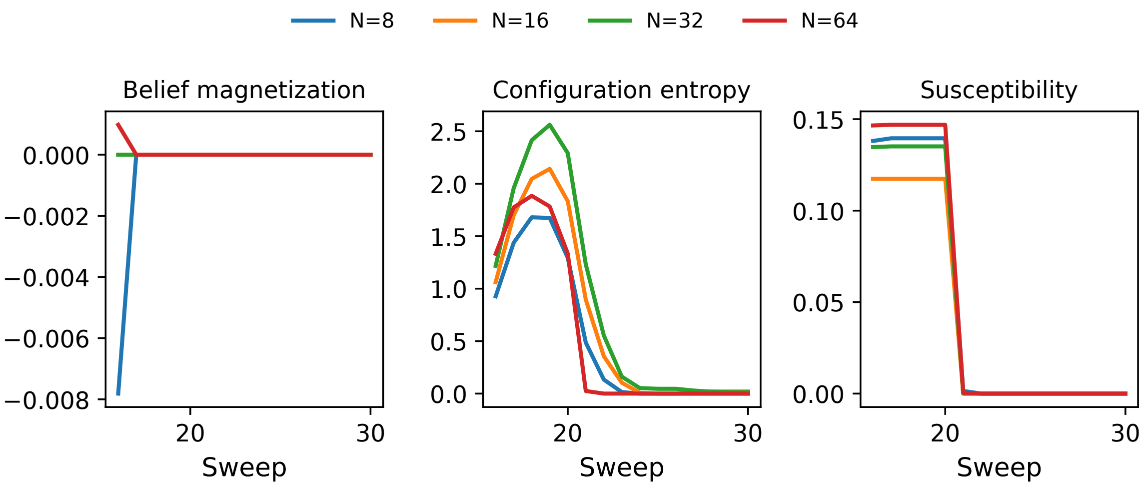

Figure 13: Direct $N = 16$ anchors in a denser inexpensive effective-model size context. These are CPU kinetic-surrogate quench responses, not direct-LLM finite-size scaling.

The Binder statistic is likewise calculated trajectory first and describes the shape of a finite $N = 16$ magnetization distribution. The source-data table retains full-phase, early-half, and late-half estimates under both cluster-mean and pooled-moment constructions; near-zero second-moment cases remain undefined. Empirical magnetization occupancy is shown alongside the Binder values so that aggregation-sensitive cumulants remain connected to the finite-window distributions from which they are calculated.

The autocorrelation sum is not an equilibrium correlation time, and these short, potentially nonstationary windows cannot establish critical slowing down. The Binder values are not a crossing analysis and do not establish a critical point or phase transition. Figure 14 therefore presents persistence, Binder, and occupancy together rather than as an additional confirmatory hypothesis family.

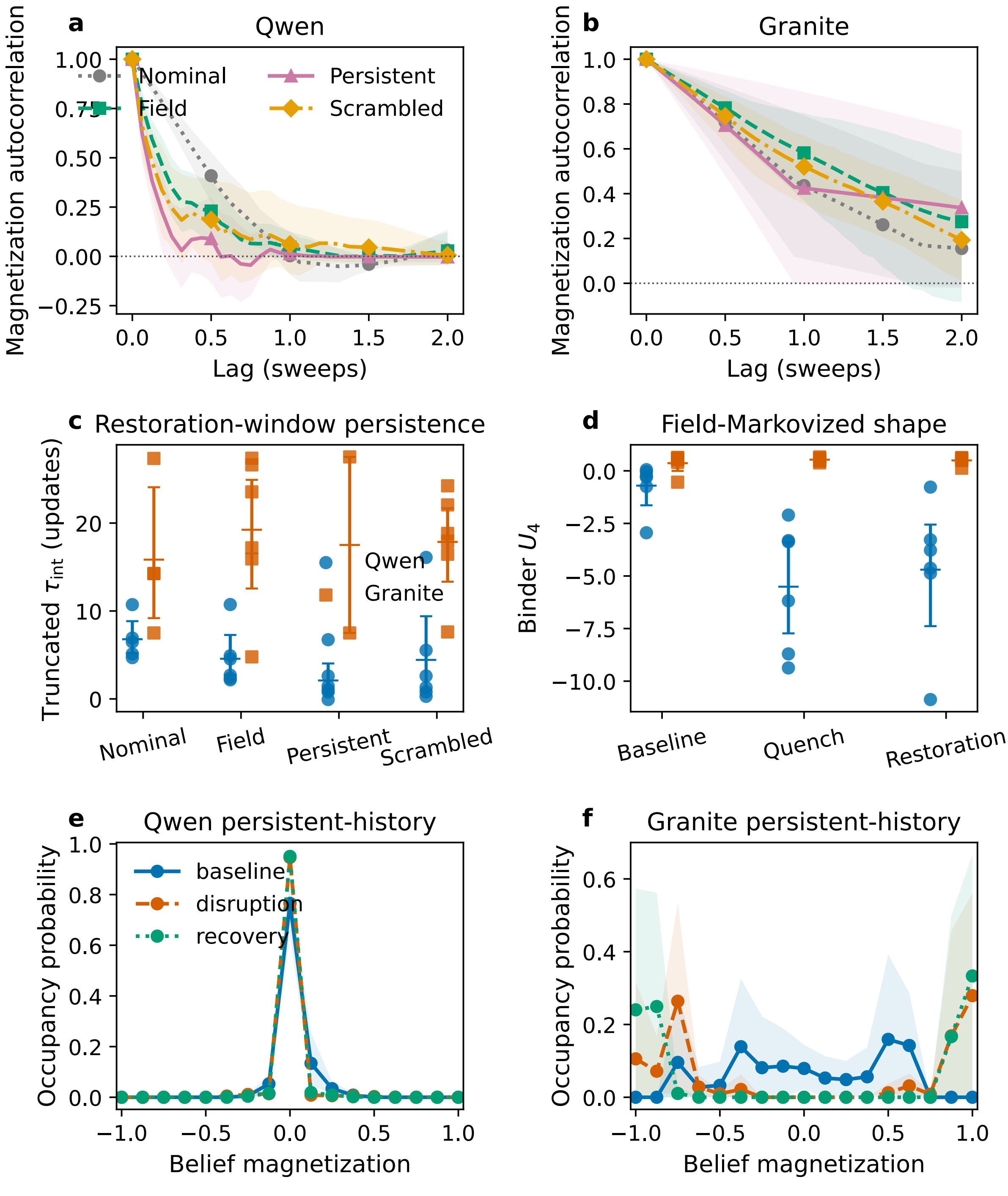

Figure 14: Finite-window persistence and order-parameter shape. Autocorrelation curves and their 95% intervals compare the nominal late window with the three field arms using complete clusters; integrated correlation sums show all cluster values at the fixed two-sweep truncation. Binder cumulants are paired with empirical belief-magnetization occupancies so their finite-sample meaning remains visible. These diagnostics do not establish critical slowing down or a phase transition.

D Computational implementation and validation

An agent receives only its own persistent state, the delivered neighbor packets available at its scheduled update, and the condition-specific prompt content defined in Section 2. Its response changes the scheduled instance's recorded belief, action, commitment, memory code, signal, and typed tool consequence; the scheduler validates but does not replace those fields. Transition records include the scheduled identity, recipient, packet bytes, and prompt and output hashes. Dedicated checks confirm that perturbing one instance's memory or decision provider leaves unrelated agents' private state unchanged.

Every experimental arm instantiates a fresh network state from its prespecified panel seeds. Model weights and the tokenizer are shared read-only for throughput, but conversational histories, generation caches, mutable agent objects, and scientific random-number state are not carried between arms. Each generation resets its per-decision seed. The two pinned model revisions reported in Section 4.2 use Transformers 4.55.4, PyTorch 2.8.0+cu128, bitsandbytes 0.47.0, NF4 double quantization, and BF16 computation under the common decoding settings given there.

Independent reconstruction validation covered all 48 trajectories and 34,560 retained decisions, with zero replay mismatches. Reconstructed aggregate tables agree with the committed reference within an explicitly audited binary64 cross-platform tolerance, not a scientific effect-size tolerance. The repository links the reported results to the model revisions, configurations, graph and seed manifests, recorded control schedules, aggregate tables, figure source data, and reproduction scripts. The original external raw tree is unavailable, so historical raw-file digest identity cannot be checked; this limitation is stated separately from the replay and aggregate comparisons that can be reproduced.

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